Name 1:_____

Name 2:_____

1. Suppose $f(x) \geq 0$ on an interval containing a, but perhaps not at a itself and that $\lim_{x \to a} f(x)$ exists. Prove that $\lim_{x \to a} f(x) \geq 0$. (*Hint*: Let $L = \lim_{x \to a} f(x)$. Assume that $L < 0$ and derive a contradiction.) A similar result holds if $f(x) \leq 0$ on this interval.

2. Use the result of Question 1 to prove that if a function is increasing on an interval and f is differentiable on this interval, then $f'(x) \geq 0$ on the interval. (*Hint*: Recall that a function f is increasing on an interval if $x_1 < x_2$ implies $f(x_1) \leq f(x_2)$.) A similar result holds if f is decreasing on this interval.